Howden Project No.:	**A053751**	**1500-ME-VEN-4330-EXP-0401_C**
Project Name:	**Thacker Pass Project**	
Document Number:	**A053751-2504-2901.1**	

EXP Project No.:	**HOU-23001048-A1**
Project Title:	**Thacker Pass Project**
PO Number:	**4500001044**
Package Title:	**Main Compressor**
VDRL Code:	**2504, 2901**
Equipment Description:	**Compressor**
Equipment Tag Number(s):	**1540-GB-001**

Compressor Performance and Operating Curves

Table of Contents:

Rev B comments addressed. This document is final.

C	Revised and reissued for approval	MWD	9/16/2025
B	Revised and reissued for approval	MWD	2/18/2025
A	Issued for review and comment	MWD	1/17/2025
Rev.	Description	Prepared By	Date

OPERATING DATA

Howden order no.	**NIF00239**	**Customer**	**Lithium Nevada Corporation**
Howden machine no.	**1052235**	**Project**	**Lithium Nevada**
Howden machine type	**SF 14.0**	**Customer order no.**	

Operating data Compressor - SF 14.0

Gas: gas composition 1

Composition: O2: 20.95% / N2: 79.05% (Vol/Mol-%)

Operating point		3	4	6		
Name		Case 3 Normal - 105% Clean	Case 4 Normal - 100% Dirty	Case 6 Furutre HRS+TGS MMEs+SCR		
Gas		gas composition 1	gas composition 1	gas composition 1		
Ambient pressure, abs	inWC	341.4862	341.4862	341.4862		
Standard density (14,7 PSIa, 32°F, dry)	lb/ft³	0.08041	0.08041	0.08041		
Site level	ft	4760				
Conditions at Battery Limits						
Volume flow s.t.p. 14,7 PSIa, 32°F, dry	scfm	115301	109811	121000		
Relative humidity	%	0	0	0		
Inlet temperature	°F	150	150	150		
Inlet pressure, abs	inWC	328.4862	328.4862	328.4862		
Pressure rise	inWC	199	218	247		
Outlet pressure, abs.	inWC	527.4862	546.4862	575.4862		
Operating Data Compressor						
Control type		Speed	Speed	Speed		
Actual volume flow	acfm	177051.326	168621.114	185802.468		
Mass flow (humid)	lb/hr	556273.9	529787.2	583769		
Inlet density	lb/ft³	0.05241	0.05241	0.05241		
Inlet pressure, abs	inWC	328.4862	328.4862	328.4862		
Inlet pressure losses	inWC	0	0	0		
Pressure rise	inWC	199	218	247		
Outlet temperature	°F	253.8	259.8	273		
Outlet pressure, abs.	inWC	527.4862	546.4862	575.4862		
Outlet pressure losses	inWC	0	0	0		
Isentropic head	ftlb/lb	16494.155	17818.148	19776.908		
Compressor speed	1/min	4174	4253	4505		
Driver speed	1/min	1654	1685	1785		

OPERATING DATA

Howden order no.	**NIF00239**	**Customer**	**Lithium Nevada Corporation**
Howden machine no.	**1052235**	**Project**	**Lithium Nevada**
Howden machine type	**SF 14.0**	**Customer order no.**	

Power						
Power on coupling	hp	5640	5687	7006		
Rated driver power	hp	7778	7778	7778		

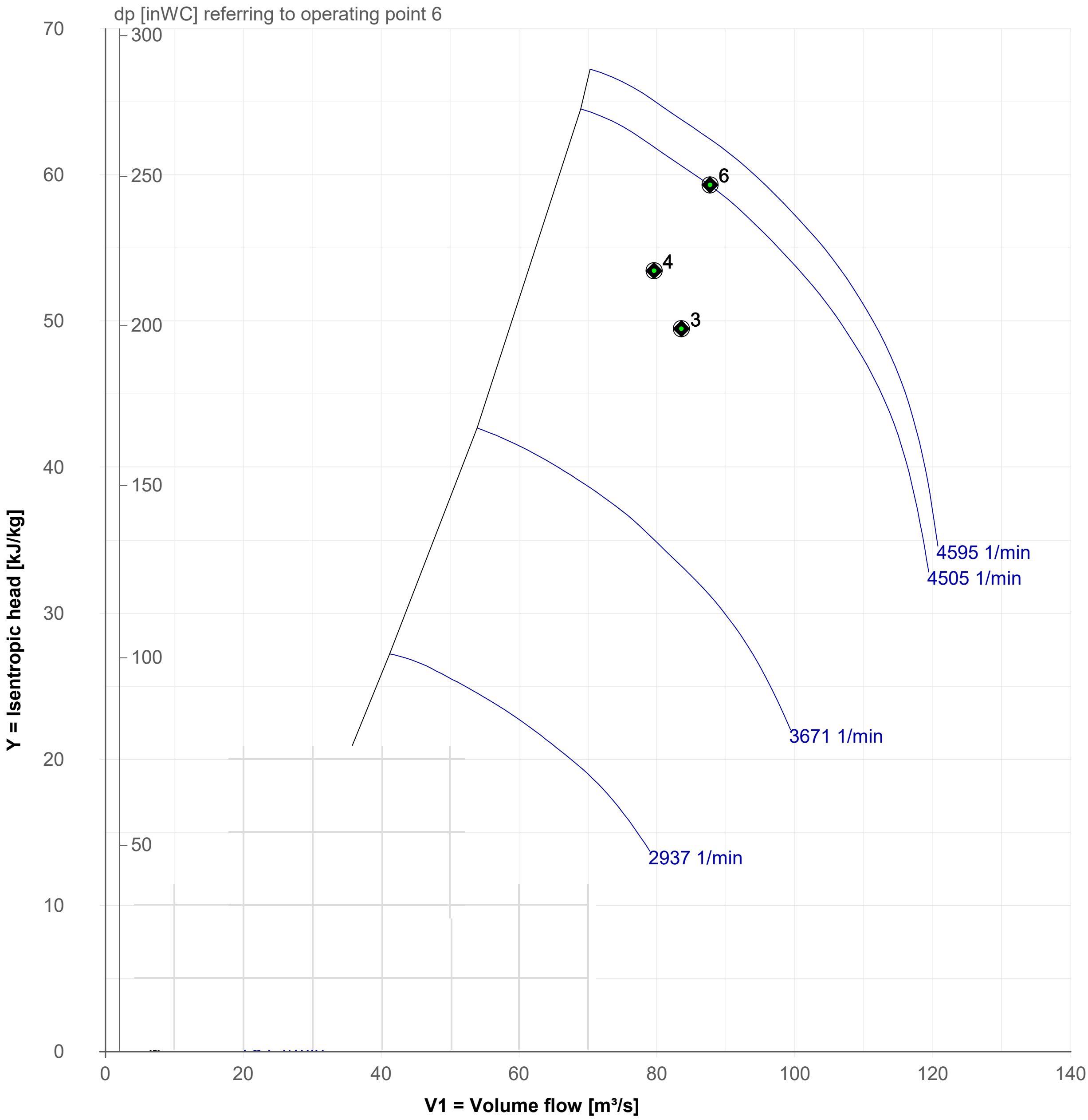
All pressure, temperature and density values are total.

Operating speed range:
n,driver: 951 rpm - 1821 rpm (102%)
n,compressor: 2400 rpm - 4595 rpm (102%)

OPERATING DATA

Howden order no.	**NIF00239**	**Customer**	**Lithium Nevada Corporation**
Howden machine no.	**1052235**	**Project**	**Lithium Nevada**
Howden machine type	**SF 14.0**	**Customer order no.**	

Performance Curve / gas composition 1 / Speed control



(): Operating points with different gas compositions. Handle with care!

Operating speed range:

n,driver:	951 rpm - 1821 rpm (102%)
n,compressor:	2400 rpm - 4595 rpm (102%)

OPERATING DATA

Howden order no.	**NIF00239**	**Customer**	**Lithium Nevada Corporation**
Howden machine no.	**1052235**	**Project**	**Lithium Nevada**
Howden machine type	**SF 14.0**	**Customer order no.**	

Operating data Compressor - SF 14.0

Gas: gas composition 1

Composition: O_2: 20.95% / N_2: 79.05% (Vol/Mol-%)

Gas: gas composition 2

Composition: O_2: 20.57% / H_2O: 1.79% / N_2: 77.64% (Vol/Mol-%)

Operating point		2	1	5	7	
Name		Case 2 Turndown	Case 1 Heat-up	Case 5 Brick Cure *)	min Case 5 Brick Cure	
Gas		gas composition 1	gas composition 2	gas composition 2	gas composition 2	
Ambient pressure, abs	inWC	341.4862	341.4862	341.4862	341.4862	
Standard density (14,7 PSIa, 32°F, dry)	lb/ft³	0.08041	0.07987	0.07987	0.07987	
Site level	ft	4760				
Conditions at Battery Limits						
Volume flow s.t.p. 14,7 PSIa, 32°F, dry	scfm	43924	27453		30000	
Relative humidity	%	0	0	0	0	
Inlet temperature	°F	150	110	110	110	
Inlet pressure, abs	inWC	338.4862	340.4862	340.4862	340.4862	
Pressure rise	inWC	56	29	2	21	
Outlet pressure, abs.	inWC	394.4862	369.4862	342.4862	361.4862	
Operating Data Compressor						
Control type		IGV (SR)	IGV (SR)	IGV (SR)	IGV (SR)	
Actual volume flow	acfm	65455.2	38001.615		41527.282	
Mass flow (humid)	lb/hr	211913	131556.4		143761.8	
Inlet density	lb/ft³	0.054	0.05775		0.05775	
Inlet pressure, abs	inWC	338.4862	340.4862	340.4862	340.4862	
Inlet pressure losses	inWC	0	0	0	0	
Pressure rise	inWC	56	29	2	21	
Outlet temperature	°F	182.8	139.9		137.5	
Outlet pressure, abs.	inWC	394.4862	369.4862	342.4862	361.4862	
Outlet pressure losses	inWC	0	0	0	0	
Isentropic head	ftlb/lb	5090.375	2530.555		1847.098	
Compressor speed	1/min	2400	2400		2400	

OPERATING DATA

Howden order no.	**NIF00239**	**Customer**	**Lithium Nevada Corporation**
Howden machine no.	**1052235**	**Project**	**Lithium Nevada**
Howden machine type	**SF 14.0**	**Customer order no.**	

Driver speed	1/min	951	951		951	
Power						
Power on coupling	hp	713	427		427	
Rated driver power	hp	7778	7778		7778	

All pressure, temperature and density values are total

*) Operating point 5 is out of performance graph and can only be operated with bypass.
Substitude point is point 7. The difference in volume flow must be recycled in the bypass.

Operating speed range:
n,driver: 951 rpm - 1821 rpm (102%)
n,compressor: 2400 rpm - 4595 rpm (102%)

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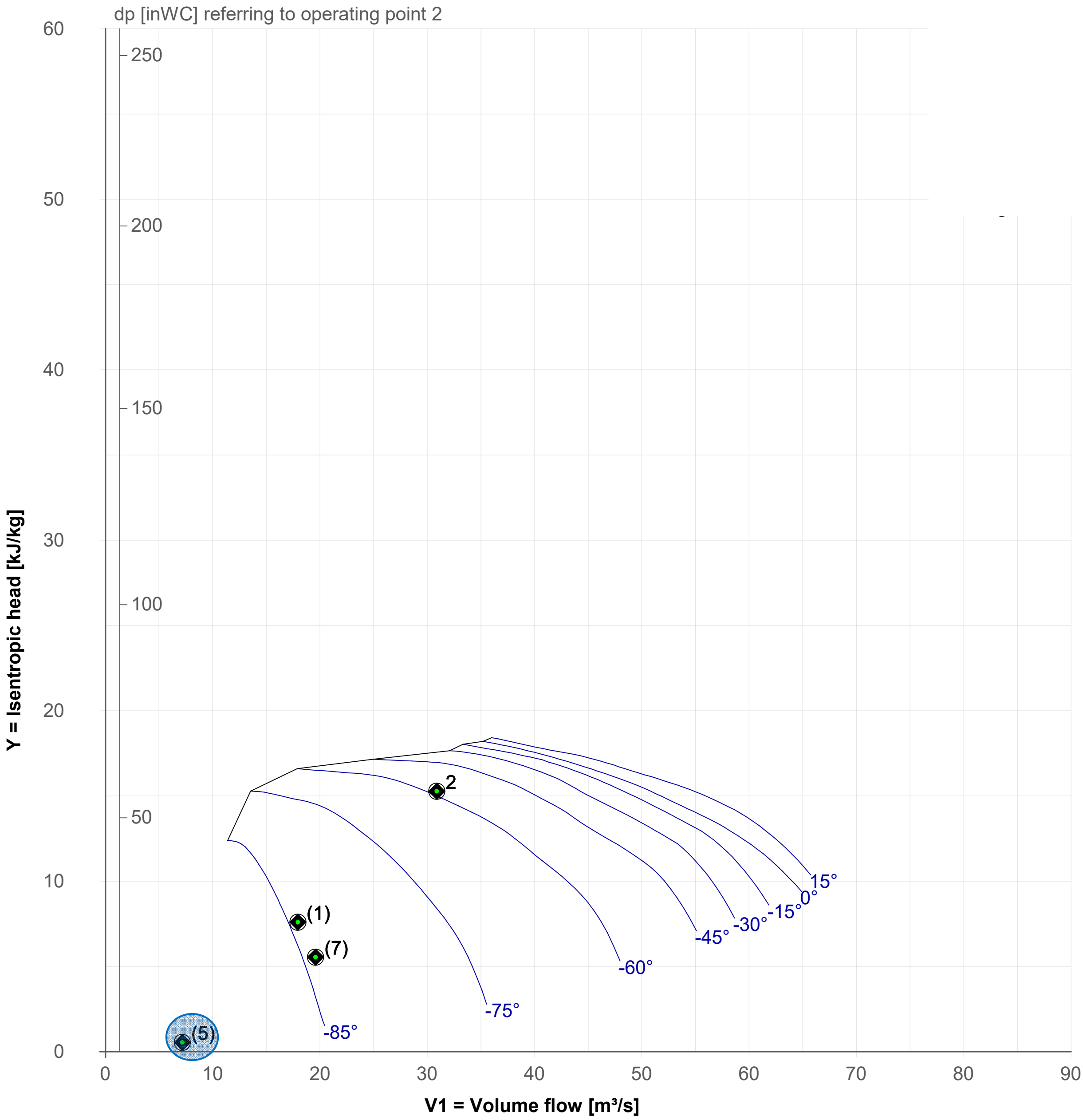
OPERATING DATA

Howden order no.	**NIF00239**	**Customer**	**Lithium Nevada Corporation**
Howden machine no.	**1052235**	**Project**	**Lithium Nevada**
Howden machine type	**SF 14.0**	**Customer order no.**	

Performance Curve / gas composition 1 / IGV (SplitRange) / n= 2400 1/min



(): Operating points with different gas compositions. Handle with care!

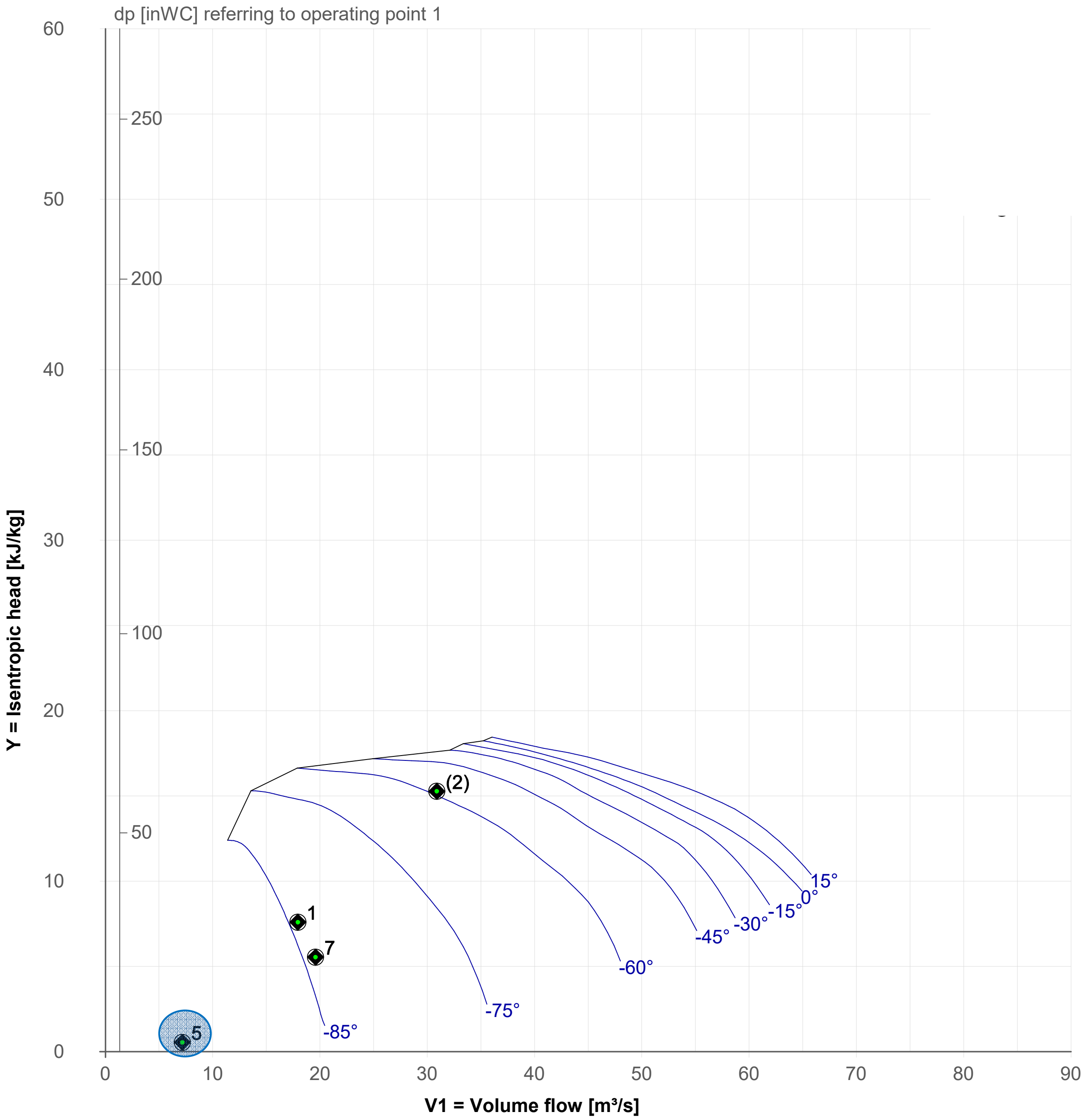
*) Operating point 5 is out of performance graph and can only be operated with bypass.
Substitude point is point 7. The difference in volume flow must be recycled in the bypass.

Operating speed range:

n,driver:	951 rpm - 1821 rpm (102%)
n,compressor:	2400 rpm - 4595 rpm (102%)

Howden order no.	**NIF00239**	**Customer**	**Lithium Nevada Corporation**
Howden machine no.	**1052235**	**Project**	**Lithium Nevada**
Howden machine type	**SF 14.0**	**Customer order no.**	

Performance Curve / gas composition 2 / IGV (SplitRange) / n= 2400 1/min



(): Operating points with different gas compositions. Handle with care!

*) Operating point 5 is out of performance graph and can only be operated with bypass.
Substitude point is point 7. The difference in volume flow must be recycled in the bypass.

Operating speed range:

n,driver:	951 rpm - 1821 rpm (102%)
n,compressor:	2400 rpm - 4595 rpm (102%)

ACOUSTIC DATA

Howden order no.	**NIF00239**	**Customer**	**Lithium Nevada Corporation**
Howden machine no.	**1052235**	**Project**	**Lithium Nevada**
Howden machine type	**SF 14.0**	**Customer order no.**	

Volume flow (in)	185802.468 acfm	Tip speed	1083.41 ft/s
Inlet temperature	150.0 °F	Temperature (out)	273.0 °F
Tot. Inlet density	0.05241 lb/ft³	Temperature out, (Max)	393.2 °F
Tot. Inlet pressure	328.4862 inWC	Density (out)	0.0764 lb/ft³
Tot. pressure rise	247 inWC	Volume flow (out)	127460.082 acfm
Speed	4505 rpm		
Input power	7006 HP		

Octave sound power level in the ducts, Lw (A)			Octave sound pressure level at a distance of 1m, Lp (A)	
Measuring surface factors:	a) Outlet:	0.0 dB	c) Machine:	20.5 dB
	b) Inlet:	0.0 dB	Pipings connected and isolated +1 dB in case of installation on a steel structure	

	32 Hz	63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz	Total
Sound power level in the duct Lw										
Outlet dB:	126.5	127	128	128	138	135.5	131	124.5	121.5	
Outlet dB(A):	87	101	112	119.5	135	135.5	132	125.5	120.5	139.5
Inlet dB:	126.5	127	128	128	140	137	132	124.5	121.5	
Inlet dB(A):	87	101	112	119.5	136.5	137	133.5	125.5	120.5	141
Sound pressure level in 1m Lp										
Compressor dB:	90	90	90	90	94	92.5	86.5	80.5	74.5	
Compressor dB(A):	50.5	64	74	81.5	91	92.5	87.5	81.5	73.5	96

Tolerances:

- Octave sound power level Lw (A) in the duct, inlet	- 6 / + 4 dB
- Total -A-sound power level Lw (A) in the duct, inlet	- 6 / + 4 dB
- Octave sound power level Lw (A) in the duct, outlet	- 6 / + 4 dB
- Total -A-sound power level Lw (A) in the duct, outlet	- 6 / + 4 dB
- Total -A-sound pressure level Lp (A) (1 m)	- 3 / + 2 dB

The levels indicated are based on measurements according to DIN 45635, the duct levels indicated are based on measurements in accordance with ISO 5136 (EN25136), with due consideration being given to the corrections C1, C2 and C3, as well as to the connections specified in section 7.2 of this standard: sound power level is a function of average sound pressure level, of the cross section surface of the measuring duct and of the product from density and sound velocity.

STARTING TORQUES

Howden order no.	NIF00239	Customer	Lithium Nevada Corporation
Howden machine no.	1052235	Project	Lithium Nevada
Howden machine type	SF 14.0	Customer order no.	

Proposed rated motor power	7778 HP
Power consumption design	7006 HP
Speed of compressor	4505 rpm
Speed of motor	1785 rpm

Speed [%]	Torque [Nm]
	Isolation valve on suction side open
0	3959 (Breakaway torque)
10	347
20	1386
30	3119
40	5545
50	8664
60	12476
70	16981
80	22180
90	26604
100	31029

The maximum torque in the table is decisive for the design of the motor and refers to the maximum inlet density rho1max = 0.06499 Mengeneinheit at the temperature T1 = 32.0 °F.

The torque changes proportional to the density and inversely proportional to the absolute temperature.

The torgue has been multiplied by a factor of 1.24; this factor results from the density ratio between the maximum density rho1max and the design density rho1tot = 0.05241 Mengeneinheit at T1 = 150.0 °F (exception: breakaway torque).

Minimum required motor mass moment of inertia JMot (WR^2) = 3999.828 $lbft^2$
Expected value of motor mass moment of inertia JMot (WR^2) > 8633.12 $lbft^2$
Mass moment inertia of the compressor, bearing, gear and coupling referred to the motor speed J (WR^2) = 48120.814 $lbft^2$